

Taehoon Kim

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Research Interests

My research aims to develop spatial intelligence for real-world, safety-critical environments. I focus on 3D Gaussian Splatting, from feed-forward reconstruction to post-hoc per-scene optimization, and on making such representations efficient enough to deploy. I am also interested in industrial and multimodal anomaly detection via distillation-based teacher–student frameworks, and in extending this notion of deviation from learned normality beyond 2D images to 3D scene representations.

Keywords: Computer Vision, 3D Reconstruction, 3D Gaussian Splatting, Efficient 3D Representation, Anomaly Detection

Education

Kyonggi University

M.S. in Computer Science

- Advisor: Junho Ahn
- GPA: 4.5 / 4.5

Sep. 2025 – Jun. 2026

Gyeonggi, Korea

Kyonggi University

B.S. in Department of Restaurant and Culinary

B.S. in Department of Artificial Intelligence – *double major*

- CGPA: 4.3 / 4.5 – Summa Cum Laude, AI major CGPA: 4.32 / 4.5

Mar. 2019 – Aug. 2025

Seoul, Korea

Gyeonggi, Korea

Experiences

V-Lab, Yonsei University

Intern

Advisor: Prof. Eunbyung Park

- Built a hybrid merge–prune compaction plug-in with per-scene optimization for feed-forward 3DGS on tttLRM (CVPR 2026); on the DL3DV-140 benchmark, matched the fully optimized baseline at 1/3 the peak VRAM.
- Extended it to MVP (CVPR 2026), a view-dependent-opacity backbone: Local Pruning with per-scene optimization rendered 4.3×/4.8× fewer primitives than MVP at 64/128 views.
- Designed the experiment infrastructure — unattended SLURM dependency chains, pre-registered decision gates, and held-out validation — running benchmark experiments spanning up to 500 GPU jobs.

Jul. 2026 – Aug. 2026

Seoul, Korea

Military–Academia Joint AI Education Program, ROK DCC

Teaching Assistant & Project Mentor

Sep. 2024 – Jul. 2026

Gyeonggi, Korea

AI Vision Lab, Kyonggi University

Undergraduate Research Assistant

Advisor: Prof. Junho Ahn

May 2024 – Aug. 2025

Gyeonggi, Korea

K-SW Square, Purdue University

Visiting Research Intern

Advisor: Eric T. Matson & Anthony Smith

Sep. 2023 – Dec. 2023

West Lafayette, IN, USA

Publications

Preprint

- [1] **Taehoon Kim**, Sehun Lee, Namgi Kim, Youjin Seo, “XSANet: Industrial X-ray Anomaly Detection via Spatial Transformer and Attention-Enhanced Teacher-Student Framework,” *paper*

International Journal

- [1] **Taehoon Kim**, Sehun Lee, Junho Ahn, “Zero-Shot Vision-Based Robust 3D Map Reconstruction and Obstacle Detection in Geometry-Deficient Room-Scale Environments,” *Computers, Materials & Continua*, vol. 86, no. 2, pp. 1–30, Dec. 2025.
- [2] Sehun Lee, **Taehoon Kim**, Sookyun Kim, Junho Ahn, Namgi Kim, “Unsupervised Structural Defect Classification via Real-Time and Noise-Robust Method in Smartphone Small Modules,” *Electronics (MDPI)*, vol. 14, no. 17, art. no. 3455, Aug. 2025.
- [3] Sehun Lee, **Taehoon Kim**, Junho Ahn, “Zero-Shot Based Spatial AI Algorithm for Up-to-Date 3D Vision Map Generations

in Highly Complex Indoor Environments,” Computers, Materials & Continua, vol. 84, no. 2, pp. 3623–3648, Jul. 2025.

International Conference

- [1] Doyoung Koh, **Taehoon Kim**, Sehun Lee, Namgi Kim, “Weighted Knowledge Distillation with DisturbLabel for Robust Learning,” 2026 17th International Conference on Ubiquitous and Future Networks (ICUFN), Milan, Italy, Jul. 2026.

Domestic Journal

- [1] Sowon Lee, **Taehoon Kim**, Junho Ahn, “A Zero-Shot Vision Algorithm for Safe Path Planning on Irregular Terrain in Unstructured Outdoor Environments,” The Journal of Korean Internet of Things Society (JKIOTS), accepted in Aug. 2026.
- [2] Doyoung Koh, **Taehoon Kim**, Junho Ahn, Namgi Kim, “A Knowledge Distillation Method Combining Prediction Confidence-based Weighting and Label Perturbation,” The Journal of Korean Institute of Information Technology (JKIIT), Vol. 24, No. 7, Jul. 2026.
- [3] Yunjeong Choi, Sehun Lee, **Taehoon Kim**, Junho Ahn, “A Robust Zero-Shot Traversable Area Segmentation Algorithm for Safe Navigation of Indoor Mobile Robots,” The Journal of Korean Institute of Information Technology (JKIIT), Vol. 24, No. 7, Jul. 2026.
- [4] **Taehoon Kim**, Jiyoung Na, Jiwon Yoon, Sehun Lee, Junho Ahn, “Multiple Camera-Based Real-Time Long Queue Vision Algorithm for Public Safety and Efficiency,” Journal of the Korea Society of Computer and Information (JKSCI), vol. 29, no. 10, pp. 47–57, Oct. 2024.

Domestic Conference

- [1] **Taehoon Kim**, Sehun Lee, Junho Ahn, “Industrial X-ray UniNet: A Robust Spatial Transformation Architecture for X-ray Anomaly Detection in Smartphone Actuator Modules,” Proceedings of the 2025 KAD Winter Conference (**Outstanding Paper Award**)
- [2] Sehun Lee, **Taehoon Kim**, Junho Ahn, “Genetic Neural Networks: Efficient Training Initialization Strategy via Weight Recombination,” Proceedings of the 2025 KAD Winter Conference (**Best Paper Award**)
- [3] **Taehoon Kim**, Sehun Lee, SeungBeom Lee, Junho Ahn, “0-VISTA: Zero-Shot Spatial AI for 3D Mapping and Detection in Complex Indoor Environments,” Proceedings of the 2025 KAD Spring Conference
- [4] Sehun Lee, **Taehoon Kim**, Doyeon Kim, Junho Ahn, “Latent Hotspot Crossover for Disentangled and Interpretable Representation Learning in Variational Autoencoders,” Proceedings of the 2025 KAD Spring Conference (**Outstanding Paper Award**)
- [5] Doyeon Kim, SeungBeom Lee, **Taehoon Kim**, Junho Ahn, “Detection of Movable Object Layout Changes in Shared-Use Facilities,” Proceedings of the 2025 KAD Spring Conference
- [6] **Taehoon Kim**, Jiyoung Na, Sehun Lee, Jiwon Yoon, Junho Ahn, “Deep Learning Vision-Based Line Recognition Algorithm Suggestion for Real-Time Intercity Bus Passenger Queue Efficiency,” Proceedings of the 2024 KSII Spring Conference
- [7] Yechan Lee, **Taehoon Kim**, Jiyoung Na, Jiwon Yoon, Sehun Lee, Hanyong Lee, “Analysis of Autonomous Driving Algorithms for Road Driving of Delivery Robots,” Proceedings of the 2023 KIIT Summer Conference

Patent

Device and Method for Determining Defect of OIS Actuator	Jun. 2026
Korea Patent Application	No. 10-2026-0116731
Method and Apparatus for Indoor Space Recognition	Jun. 2026
Korea Patent Application	No. 10-2026-0116730
Deep Learning-Based Queue Recognition System	Oct. 2024
Korea Patent Application	No. 10-2024-0144283

Projects

Pedestrian Safety System for Large-Scale Buildings Using 3D Congestion Mapping with Collaborative Robots and Vision AI for Disaster Response NRF	Sep. 2025 – Aug. 2026
Acoustic-Based Real-Time Drone Localization & Tracking AI System ITSA & ROK DCC	Jul. 2025 – Dec. 2025
Improvement of Detection and Classification Capabilities in SUB2 ASS'Y GAP Inspection im co., ltd.	Nov. 2024 – Feb. 2025
Real-Time Multimodal Vision-Integrated Military Fitness Assessment ITSA & ROK DCC	Sep. 2024 – Dec. 2024
Improvement of Detection and Classification Capabilities in X-RAY Ball Inspection im co., ltd.	Jul. 2024 – Aug. 2024
Enhancing Culinary Experience: Real-Time Interactive Cooking Assistance Utilizing Mixed Reality Purdue University	Sep. 2023 – Dec. 2023

Awards

Best Paper Award , 2025 Korea Data Science Winter Conference	Dec. 2025
Outstanding Paper Award , 2025 Korea Data Science Winter Conference	Dec. 2025
Outstanding Paper Award , 2025 Korea Data Science Spring Conference	Jun. 2025
Merit Award , Kyonggi University SW Imagination Company Project Competition	Nov. 2024
Gold Award , Kyonggi University Advanced Capstone Design Project Competition	Jun. 2024
Gold Award , Kyonggi University Foundational Capstone Design Project Competition	Jun. 2023
Grand Award , Kyonggi University GamsungSW IT Competency Competition	Dec. 2019

Honors

Letter of Appreciation , ROK Defense Communication Command · Award for outstanding contribution in AI development and mentoring	Apr. 2026
AI & SW Talent Scholarship , Noh Youngbaek Scholarship Foundation · Award for excellence in the AI & Software field	Nov. 2025
Faculty Council Honors Scholarship , Kyonggi University · Award for outstanding academic achievement	Jun. 2025
Highest Honors , Kyonggi University · Graduate 1st in Dept. of Restaurant & Culinary	2025
Semester High Honors , Kyonggi University · Award for Academic Excellence (1st)	2019, 2022 · 2023 (1st Sem), 2024

Technical Skills

Programming languages

Python, C, Java, R

Frameworks & Libraries

PyTorch, TensorFlow, Keras, NumPy

Tools

CUDA, Linux, SLURM, Git, gsplat

Language Skills

TOEIC – Total Score 920	Aug. 2024
TOEIC Writing – Advanced Mid (AM)	Apr. 2024
TOEIC Speaking – Intermediate High (IH)	Apr. 2024
Official English Presentation as University Representative	Jul. 2024
Kyonggi University National Center of Excellence in Software · Presented program highlights and achievements to a CHED delegation, Philippine President's office	news article
English Interpretation & Translation Volunteer	Oct. 2023
Purdue Military Research Institute (PMRI) & IITP · Interpreted AI lectures and supported communication for ROK ROTC officers	